

Skills

Languages: C, C++, Python, Bash, VHDL, XML

Tools & Utilities: Linux, Git, Jira, CANalyzer, MATLAB/Simulink

Libraries: NumPy, Pandas, Scikit-Learn, Pytest, OpenCV, Threading

Experience

Autonomous Software Lead | UWaterloo EcoCAR Team

May 2025 - Present

- Deployed a Linux-based Raspberry Pi **CAN** gateway to Cadillac LYRIQ ECUs; built Bash tooling and a multithreaded **Python** service to drive status lights/switch inputs with low-latency messaging, tested with HIL.
- Built a 2D **LiDAR** object-detection pipeline with Python, using **C++** for sensor I/O via TCP; implemented machine learning models using scikit-learn and NumPy, with OpenCV for data visualizations, achieving **96%** detection rate.
- Deployed and tested vehicle software on an **Ubuntu** ADAS computer via terminal bring-up, utilizing **CANalyzer** to troubleshoot in-vehicle **CAN buses**.
- Designed radar + camera sensor fusion with Kalman filtering in **Simulink** with RTMaps middleware and Python, improving object-tracking accuracy by **85%** for perception in ADAS features.

Technology Product Manager Co-op | Purolator

Jan 2026 - April 2026

- Performed data-driven partner targeting using **Python** (pandas), analyzing **50+** eCommerce partners and conducting a competitive landscape review to prioritize revenue opportunities.
- Architected a centralized product tracking system, developing an **API-driven AI automation agent** that streamlined the intake process and accelerated time-to-triage by **70%**.
- Drove product MVP from discovery to engineering-ready scope, defining system constraints/dependencies, and tracking execution in **Jira** across cross-functional teams.

Software QA Engineer Co-op | i4i

Jan 2024 - Dec 2024

- Engineered an internal automation tool in **Python** and XML to parse Excel metadata and dynamically generate Word documents, increasing conversion team productivity by **>90%**.
- Developed and maintained unit test suites using **Pytest**, automating **functional** and **regression** testing while also leading a team in executing performance tests for multiple enterprise customers.

Projects

International Finalist - LiDAR Safety System | Enhanced Safety in Vehicles SSTDC

- Selected as an international finalist by the **NHTSA** and **Transport Canada** for the Enhanced Safety of Vehicles conference to present a 2D LiDAR pedestrian awareness safety system module that provides vision in blind spots.
- Leveraged **C++** to interface with the LiDAR over **ethernet**; created a data pipeline with **Python** using scikit-learn and NumPy to detect pedestrians by leveraging machine learning techniques.
- Delivered a deployable pedestrian perception system that communicates with the vehicle over **CAN** networks to trigger driver alerts, passing HIL and VIL testing with a **>96%** detection rate while aligning with **ISO 26262** principles.

Autonomous Taxi Hackathon Winner | Hack the Move

- Secured 1st place by engineering an end-to-end **autonomous taxi simulation** that successfully integrated perception, localization, planning, and control systems in MATLAB and Simulink.
- Engineered a deep-learning perception pipeline in Python utilizing **YOLOv8** to detect stop signs and traffic lights with **88% accuracy**.
- Implemented a **finite-state controller** for stop-and-go maneuvers with precise target-speed profiling, validated by achieving a 100% collision-free record and zero safety infractions across varied simulated urban scenarios.

Education

University of Waterloo

BASc. Mechatronics Engineering – GPA: 93%

Sept 2023 - Apr 2028